



WHITE PAPER

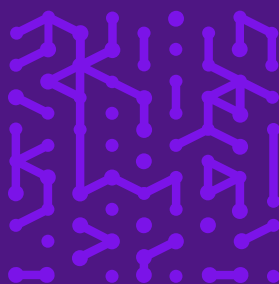
Medical and IoT Device Security for Healthcare

Managing Risk and Ensuring Patient Safety with 21st Century Healthcare

Overview

Modern healthcare environments have undergone extensive digital transformation, introducing many connected devices, including Internet of Things (IoT) and Internet of Medical Things (IoMT) technology. The new breed of connected medical devices brings the promise of improved patient care, better clinical data, improved efficiency, and reduced costs—but they also bring increased security risks. According to Armis Labs, 51% of healthcare delivery organizations reported a cyberwarfare incident in 2023, with a notable risk related to unmanaged IoT devices.¹

This white paper will examine the security risks of medical devices that affect healthcare delivery organizations and the patients that they serve. We will also explore a solution to the security problem—one that encompasses not just biomedical devices, but all the various “smart” devices that are present in healthcare delivery organizations that add to the risk.



Medical Device Exposure Risks in Healthcare

Connected medical devices offer the potential to improve patient care and operational efficiency, but they also introduce three distinct security challenges:

Lack of Visibility and Inventory Capabilities

An accurate inventory forms the basis for any security or risk program. The problem is that medical devices and other smart devices in a healthcare ecosystem are difficult to discover and inventory. Since unmanaged IoMT, IoT, and smart devices don't support inventory agents and are often missed by network discovery scans, security teams relying on traditional inventory methods struggle to get a clear understanding of their true risk.

Even when biomedical teams have an inventory of the devices their departments support, usually in the form of spreadsheets, the inventories are often outdated and require resource-intensive efforts to update. Moreover, the device list is limited to what the biomedical team manages and wouldn't include other devices, such as those managed by the diagnostic imaging or diabetes teams. The challenges are further complicated by the fact that different departments and teams often purchase devices and other innovative technologies in a decentralized manner independent of the security team, creating potentially enormous blind spots.

Inherent Security Control Limitations

Beyond asset visibility, there are limitations on the security controls that can be applied to medical devices. While traditional enterprise devices, such as laptops and servers, support a host of traditional security tools, including inventory and patching agents, medical devices simply cannot. For example, many devices, such as Alaris PCU pumps run proprietary operating systems (OSs) that do not support agents.

In other cases, despite running a mainstream OS, such as Microsoft Windows, the devices are certified by the vendor with specific configuration parameters. In these cases, relying on traditional agent-based security tools, or even installing native Windows security patches, can result in unresponsiveness or unexpected behavior that can impact patient safety. This inability to patch or upgrade has resulted in millions of devices in healthcare environments running decades-old legacy OSs and vulnerable software, further amplifying risks to patients and organizations.

The Armis advantage

Complete visibility

- Powerful discovery
- Unified asset inventory

Contextual intelligence

- Multidimensional views
- Analytics and intelligence

Continuous security

- Vulnerability prioritization and remediation
- Early warnings of potential threats

Rapid time to value

- Modern cloud architecture
- Industry leader, trusted partner

Inability to Contextualize Clinical and Device Risk

Healthcare device ecosystems are full of unmanaged devices. And with different departments using an array of new device types, understanding their unique contexts is critical to the risk assessment process. And it's also impossible to achieve through the traditional security software stack. For example, although vulnerability scanners can identify common vulnerabilities and exposures, they treat each endpoint as the same (a compute device running vulnerable OS and software).

In a clinical environment, however, a legacy workstation embedded into a large MRI scanner actively providing patient care poses a significantly higher risk compared to the same OS with the same vulnerabilities on a workstation found in a non-clinical environment. Manufacturing Disclosure Statements for Medical Device Security (MDS²) files also provide an abundance of privacy and security functions of the device that healthcare organizations must take into consideration.

Proven Value for HDOs

Armis Centrix™, the cyber exposure management & security platform, delivers tangible value via proactive cybersecurity for healthcare delivery organizations, including the following benefits.

Protect patient safety

Alert security managers to cyberattacks and compromised devices, including medical, clinical, and smart devices of all types. The ability to complete contextualized risk analysis, such as understanding clinical risk related to FDA recalls, allows teams to prioritize securing the most critical at-risk patient care devices.

Maintaining availability of patient care services

Proactively monitor for network issues and manage risk across all devices that can impact patient care service. Enable enhanced monitoring and real-time alerting for the most critical clinical devices.

Protecting patient privacy

Monitor for the various ways patient privacy is put at risk, from detection of external transfer of unencrypted PHI, to clinical system service account compromise, or compromised IP cameras streaming out to the internet.

Enable patient care innovation safely and securely

Empower research and clinical departments to improve patient care delivery through the secure adoption of innovative technologies and migration from legacy methods and systems.

Ensure audit and regulatory compliance

Streamline auditing of systems and ensure compliance with regulations, such as HIPAA or PCI-DSS for payment systems, and various audit requirements.

Automate device inventory

Reduce person-hours focused on tracking devices with a dynamically updated real-time view of all hardware and software assets in the environment.

Reduce costs with data-driven purchasing decisions

Generate a complete, real-time inventory of all medical and other devices, find lost equipment, more efficiently utilize existing equipment, and leverage utilization analytics to justify purchasing additional equipment, ultimately reducing cost of over-investment in equipment.

See and stop ransomware attacks

Identify WannaCry and other ransomware in real time, reducing the risk of operational downtime and reputational damage.

Deliver faster vulnerability and threat detection

Real-time passive identification of vulnerabilities and detection of threats ensures attack surface management and faster time to response for incidents, minimizing the risk and impacts of cyberattacks.

Streamline incident response

Fine-tune SOC workflows and achieve faster response through automated enforcement of security controls in the event of an incident. Automatically contain and remediate cyber threats as they happen in real-time, reducing overall operational risk and downtime.

Ultimately, traditional security tools simply do not capture the true contextualized device risk, including the behavioral and clinical factors. And this leads to an inaccurate quantification of a healthcare organization's true security posture and risk.

It's Not Just Medical Devices That Impact Patient Care

Medical devices like the ones described above are not the only unmanaged devices that pose a risk. They are also not the only devices that can directly impact patient care. Just consider a few parts of the ecosystem that supports the patient journey:

- Check-in kiosks
- Tablets and iPads
- Handheld scanners
- Security cameras
- Digital signage displays
- Elevator control systems
- Smart thermostats
- Smart TVs
- HVAC systems controlling humidity and temperature
- Porter communication systems
- And more

Malfunction or downtime of building management systems, such as the HVAC units that regulate humidity levels in an operating room, can result in canceled surgeries. This is a prime example of a non-medical device that directly impacts patient care. The implications are striking when you consider that 64 percent of healthcare delivery organizations estimate that at least half of all devices on their network are unmanaged or IoT devices, including medical devices.² While hospital and security teams tend to focus on medical device security, it is imperative to 'zoom out' and include other IoT and smart devices as part of the risk assessment scope.



Healthcare is a Frequent Target

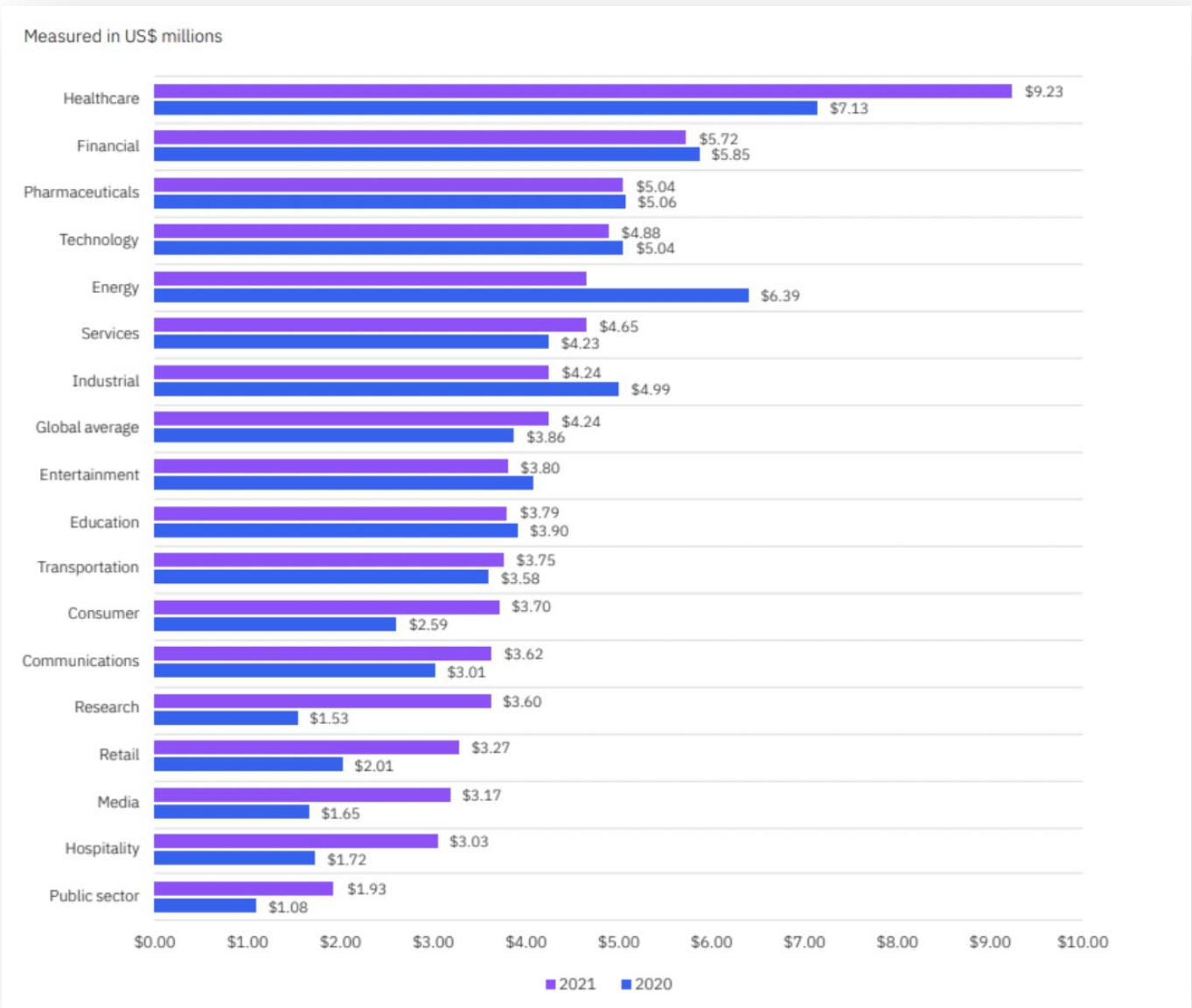
Statistics show that healthcare delivery organizations are among hackers' favorite targets. Why? Because medical records contain information that can be used for identity theft. As a result, the resale price for a healthcare record is orders of magnitude more than the resale price of a stolen credit card number.

Since hacking healthcare organizations is now so lucrative, the number of security breaches experienced by healthcare institutions has skyrocketed. And all it takes is a quick Internet search to see that healthcare is top target for cybercriminals year after year. In fact, the HIPAA Journal reported that 2024 saw the largest single healthcare data breach of all time with over 100,000,000 records compromised.³

**Over 93 million records were compromised
in healthcare data breaches in 2023.**

– HIPAA Journal⁴

To make matters worse, data breaches are more costly for healthcare providers than for any other type of business. This is due to the stringent penalties and costs that are mandated by HIPAA regulations. According to the Ponemon Institute, the average cost of a data breach for healthcare providers was more than \$9.23million.⁵



Average cost of a data breach by industry.⁶

Securing Medical Devices and More

Patient monitoring devices

- **Medical devices** – Smart medical devices, infusion pumps, ventilators, incubators, telemetry, smart stethoscopes, medical imaging
- **Clinical monitors** – Electrocardiogram (ECG), heart rate, pulse oximetry, ventilators, capnography monitors, depth of consciousness monitors, regional oximetry, biopatch technology, and respiratory rate
- **Smart patient room** – Smart beds, hand hygiene, fall detection
- **Virtual care and telemedicine** – Remote health telemetry and care delivery

Patient experience devices

- **Registration devices** – Kiosks, label printers, tablets, and pagers
- **Digital signage** – Smart TVs, wait time screens, electronic whiteboards
- **Patient transport** – Elevator systems, porter communication devices, lab specimen transport

Building management system monitoring devices

- **Emergency response** – Code blue infrastructure, fire alarms and suppression, dedicated telephony, helipad lightings and control
- **Security** – Video surveillance, door locks and entry systems, security communication systems
- **Utilities and power** – Power monitoring, emergency generators, power distribution, water supply and quality monitors
- **Environmental controls** – HVAC, lighting, room control, humidity monitoring, tissue and blood refrigerators

Remote wellness and chronic disease monitoring devices

- **Implantable devices** – Pacemakers, defibrillators, neurostimulators
- **Wearables** – Wristbands, biopatches, smartwatches, ear buds
- **Remote clinical monitors** – Spirometer, pulse oximeter, ECG, glucometer, fall detection

From Hacking Patient Data to Hacking Patient Care

Cyber attackers continue to expand their focus and are no longer just concentrating on extracting healthcare records and patient data. For example, ransomware gangs have focused on gaining control over medical devices and demanding payment for returning control. And their financial success has resulted in increasingly frequent and costly cyberattacks and constant news headlines over the past several years. In 2023, for example, 725 breaches were reported to OCR and exposed or impermissibly disclosed more than 133 million records.⁷ And the threats to healthcare organizations persist. In 2023, the healthcare industry saw the most ransomware attacks of all critical infrastructure sectors, according to the FBI.⁸

Today, given their inherent vulnerabilities, attackers are increasingly looking to medical devices and other unmanaged assets as onramps for ransomware and other types of attacks. To understand why, just consider a few vulnerabilities discovered by the security community.

TLStorm

In spring 2022, Armis researchers discovered a set of three critical vulnerabilities in APC Smart-UPS devices, a widely used emergency backup power source in healthcare environments. Attackers could exploit the vulnerabilities in numerous ways. For example:

- Installing malicious firmware on a device to establish a stronghold for more attacks
- Remotely altering the power supply to sensitive medical assets connected to the device
- Bridging the cyber and physical realm by causing devices to go up in smoke and cause physical damages.
- Initiating a ransomware attack, threatening to take control of the backup power supply to a data center supporting critical ICU applications, for example.⁹

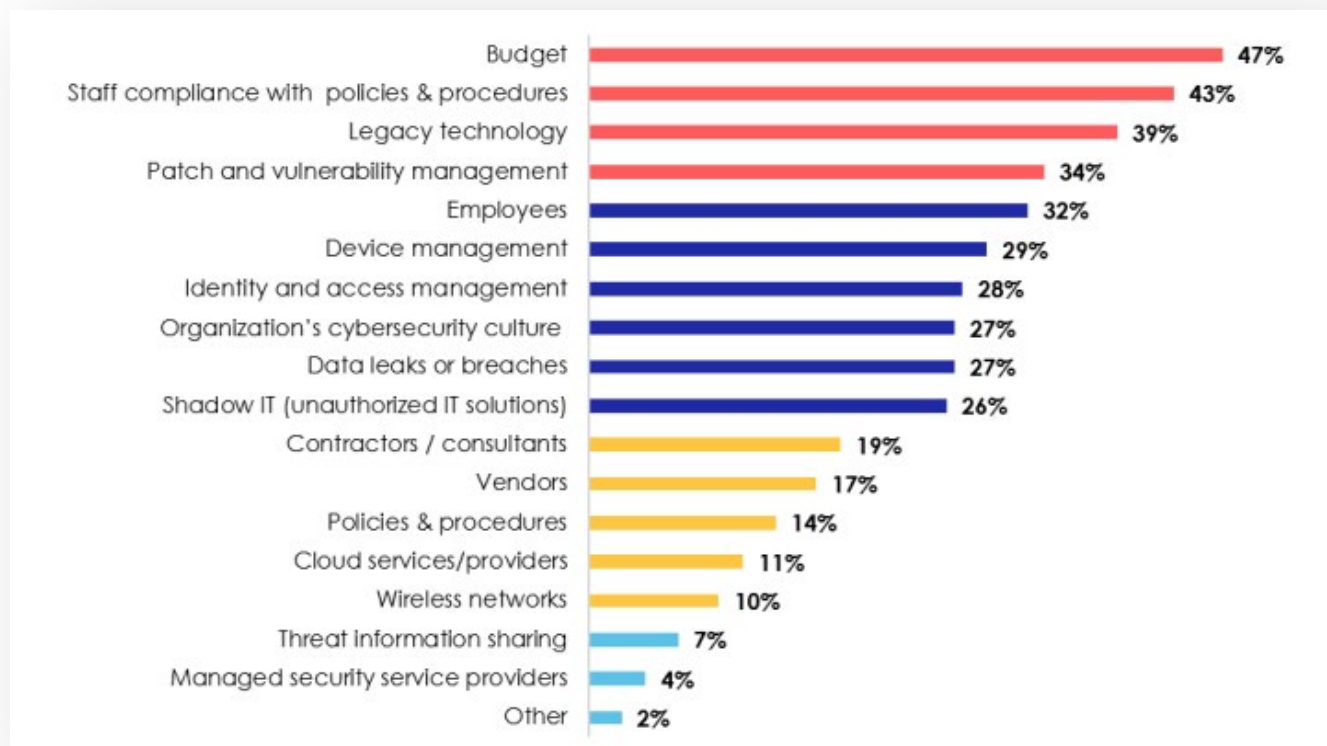
Log4j vulnerabilities

In January 2022, the U.S Department of Health and Human Services (HHS) Cybersecurity Program warned health organizations about Log4j vulnerabilities, noting that “health sector adversaries are actively leveraging these vulnerabilities” and that “vulnerabilities in ubiquitous apps will present similar issues in the future.”¹⁰

URGENT/11

In recent years, the Armis announcement of URGENT/11 illustrates the potential unknowns and patching challenges associated with any given vulnerability. For example, in 2019 Armis demonstrated how URGENT/11 vulnerabilities allow the takeover of a patient monitor, potentially changing the readings on the device without notice. Not long after the initial announcement, Armis, the FDA, and DHS jointly announced additional medical devices at risk due to URGENT/11.¹¹ By early 2022, researchers at a third-party organization demonstrated that the majority—75 percent—of 200,000 analyzed infusion pumps still included a host of security gaps, including URGENT/11.¹² The actual number of devices in the marketplace still at risk from URGENT/11 vulnerabilities could range from hundreds of thousands to millions.

Couple the potential security holes in unmanaged assets with the fact that patch and vulnerability management and device management are among healthcare security teams biggest challenges, and it's easy to understand attackers' motivations.



The biggest security challenges facing organizations, according to the HIMSS Cybersecurity Survey Report.¹³

The new healthcare cyberattack

A typical healthcare provider will have a variety of traditional IT security tools, such as firewall, intrusion detection, endpoint security, and encryption controls as mandated by HIPAA. The organization also typically includes a variety of healthcare equipment such as:

- Blood gas analyzers
- Diagnostic equipment (for example, PET scanners, CT scanners, MRI machines)
- Therapeutic equipment (for example, infusion pumps, medical lasers, and LASIK surgical machines)
- Life support equipment (for example, heart-lung machines, medical ventilators, extracorporeal membrane oxygenation machines, and dialysis machines)
- PACS and imaging systems

For entrance into the healthcare facility network, cyber attackers usually prefer the easiest route which is either a phishing attack, or a device that is exposed directly to the Internet. In both cases, legacy and unpatched devices have security flaws that attackers can exploit for drive-by downloads and ransomware execution. And once they've compromised an asset, cyber attackers can quickly and easily move laterally through traditionally flat hospital networks and infect many other devices, including medical devices, and entire care systems.

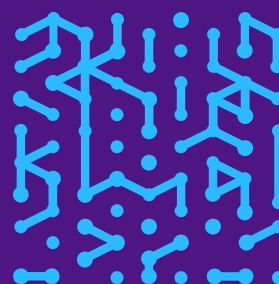
The Shodan search engine might reveal security cameras, HVAC systems, or other such devices that can be attacked. Attackers even plant rogue devices to compromise networks, such as introducing a WiFi pineapple into the lobby or nearby parking lot. Armis frequently finds pineapples in enterprise and healthcare environments even if the organization has invested heavily in security tools and personnel. Finally, there are various remote attacks that can be used to bypass or compromise the firewall, such as CDPwn, URGENT/11, or DNS rebinding.

Just take the URGENT/11 vulnerabilities announced by Armis. Companies such as GE Healthcare, Philips, Drager, Spacelabs, and BD issued advisories that their products were impacted by URGENT/11. There were 11 vulnerabilities, including six that could allow an attacker to take over a medical or other device. Armis demonstrated how an attacker could take over a patient monitor.¹⁴

Targeting hospital devices

Attackers have moved from health care systems to health care devices. There are documented cases of MRIs and CT scanners impacted by malware. Potential consequences include:

- Locked and inoperative systems
- Systems operating incorrectly
- Disruption of scan signals
- Altered results
- Altered radiation exposure



A medical device exposure

After the initial announcement of URGENT/11 vulnerabilities impacting Wind River VxWorks, hospitals using Armis identified additional medical devices with the impacted IPnet vulnerability. Armis confirmed six additional real-time operating systems were impacted (OSE by ENEA, Integrity by Green Hills, ThreadX by Microsoft, Nucleus RTOS by Mentor, ITRON by TRON Forum, and ZebOS by IP Infusion). Armis worked with the FDA, DHS, and an impacted device manufacturer, BD Alaris, to address the vulnerabilities and issue advisories. The vulnerabilities included the ability to gain entry via firewalls and simple devices like printers, as well as medical devices. The DHS recommended that hospitals minimize network exposure for all control system devices and/or systems, and ensure that they are not accessible from the Internet.¹⁵ They also recommended locating control system networks and remote devices behind firewalls, and isolating them from the business network.



The Armis solution

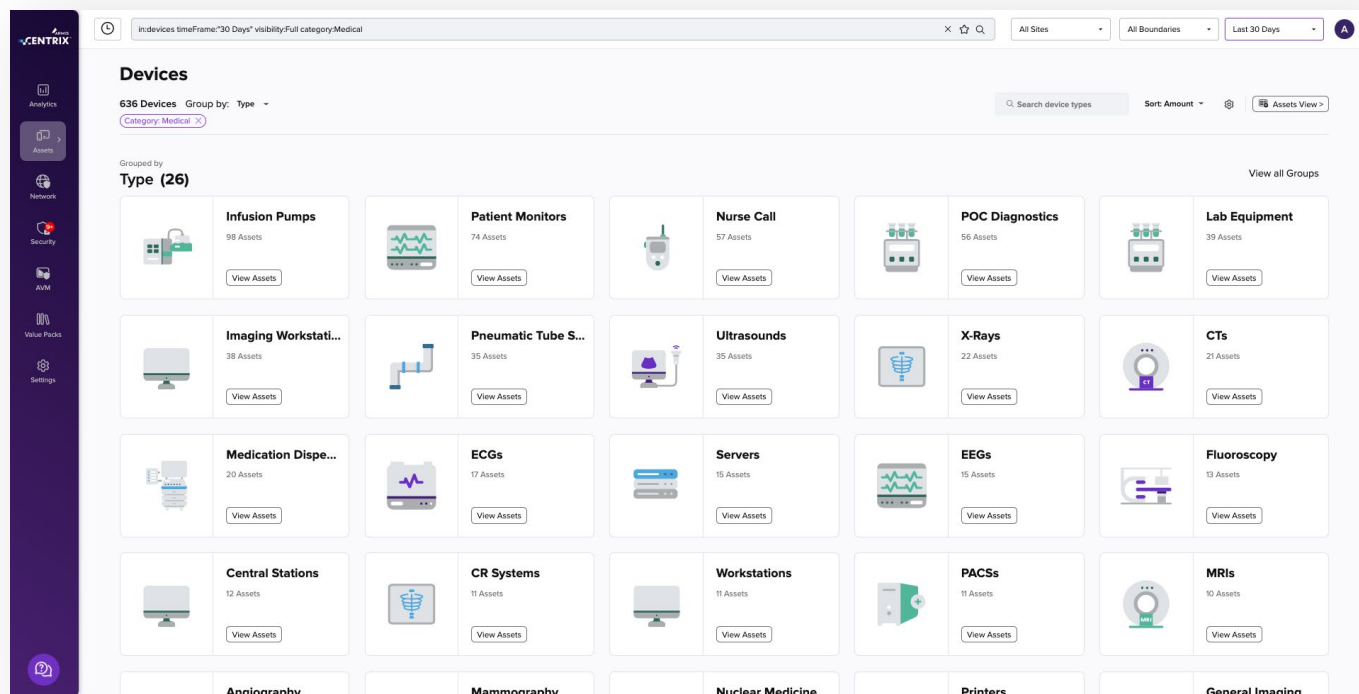
Armis is purpose built to address the need for medical and IoT device security by today's healthcare delivery organizations. Armis is an enterprise-class agentless and passive device security platform that provides three essential capabilities:

Armis is agentless and passive—critical for medical devices that can't take an agent or that you cannot scan.

Discover

Armis allows you to see all managed and unmanaged devices in your environment by type, both on your network and in your airspace. This is critical because we find that most healthcare providers are unaware of approximately 40 percent of the devices in their environment. The Armis platform operates in a completely agentless and passive manner, running without the need for software installation on devices or intrusive scanning. Armis provides visibility in near real-time, dynamically updating the inventory as new devices come online or changes are detected, ensuring teams have the most up-to-date view of their environment.

Through a simple out-of-band connection to your network, the Armis platform profiles and classifies devices, users, connections, applications, and operating systems throughout your environment. Armis shows you existing devices and the connections, including connections to unmanaged devices or rogue networks that you might not be aware of. This out-of-band connection also ensures no interruption or latency is introduced to the production network, making it a completely seamless deployment.



Example of the asset inventory by type dashboard on the Armis platform.

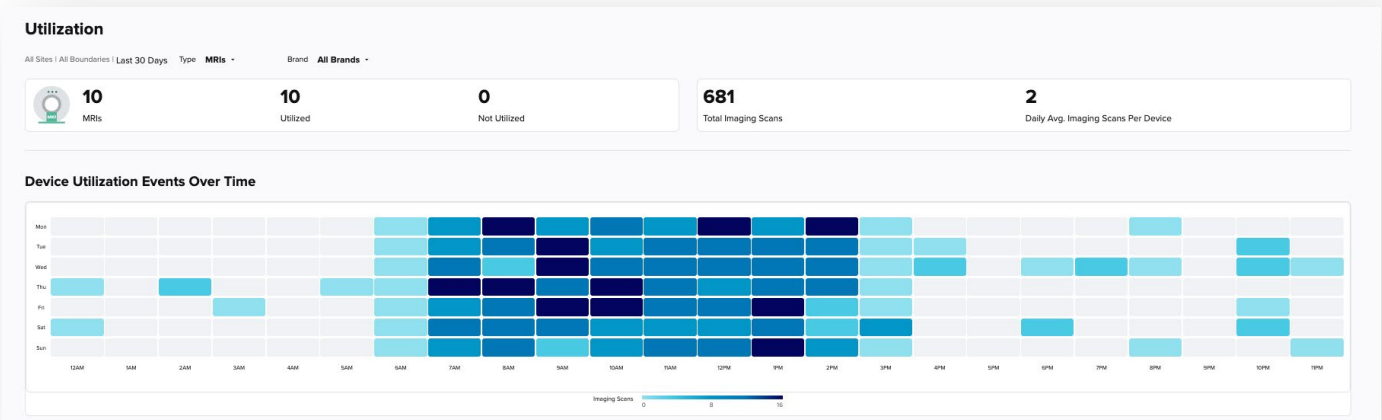
Healthcare providers are unaware of approximately 40 percent of the devices in their environment.

The Armis platform utilizes our proprietary Asset Intelligence Engine—a crowd-sourced, cloud-based knowledgebase tracking over 5 billion devices. This lets Armis accurately classify every device in your environment, including managed and unmanaged endpoints as well as non-traditional devices found in healthcare environments, such as laboratory instruments, heart monitors, infusion pumps, X-Ray systems and clinicians' handheld devices.

The comprehensive device inventory generated by Armis includes critical information like device manufacturer, model, serial number, physical location, username, operating system, installed applications, and connections made over time. Armis also detects medical device specific properties including FDA recalls, MDS² files and properties and utilization metrics.

Armis even detects medical device specific risk, such as the transmission of unencrypted Patient Health Information (PHI), MDS² files and properties and utilization metrics.

This risk score helps your security team understand your attack surface and meet compliance with regulatory frameworks (for example, the NIST framework) that require identification and prioritization of vulnerabilities.

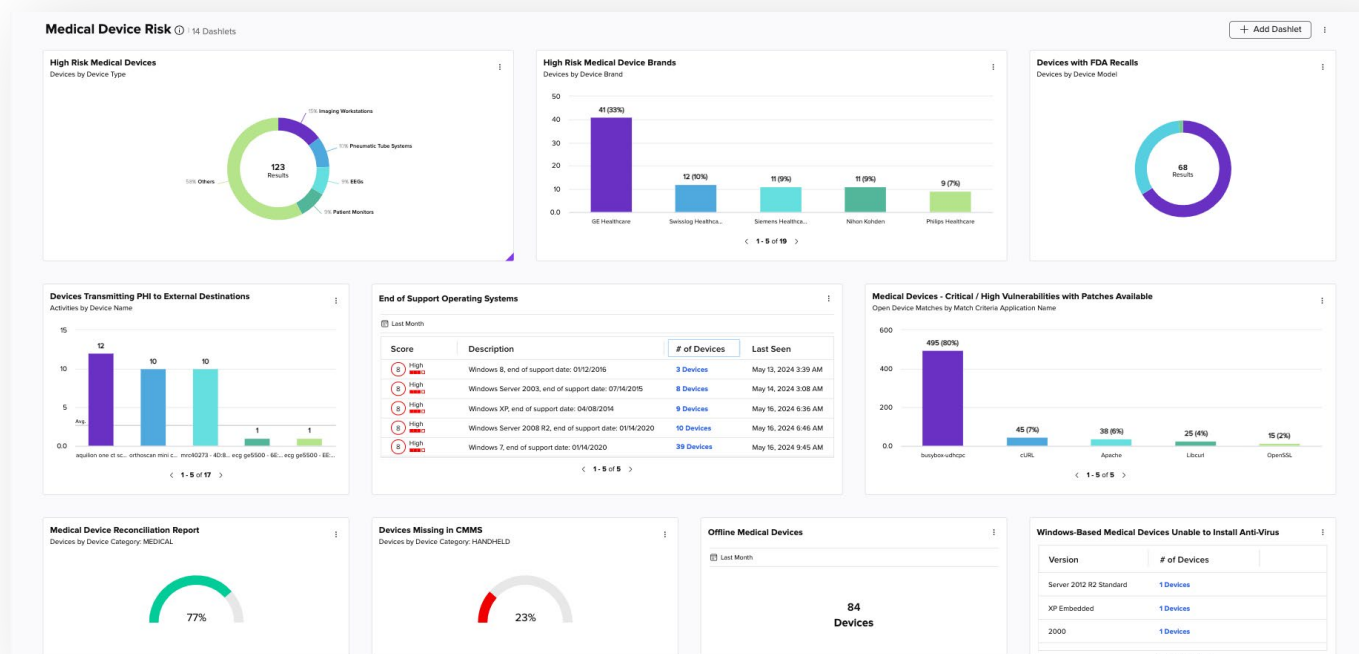


Armris Centrix™ shows you everything from the big picture of assets in your environment down to granular device views.

Analyze

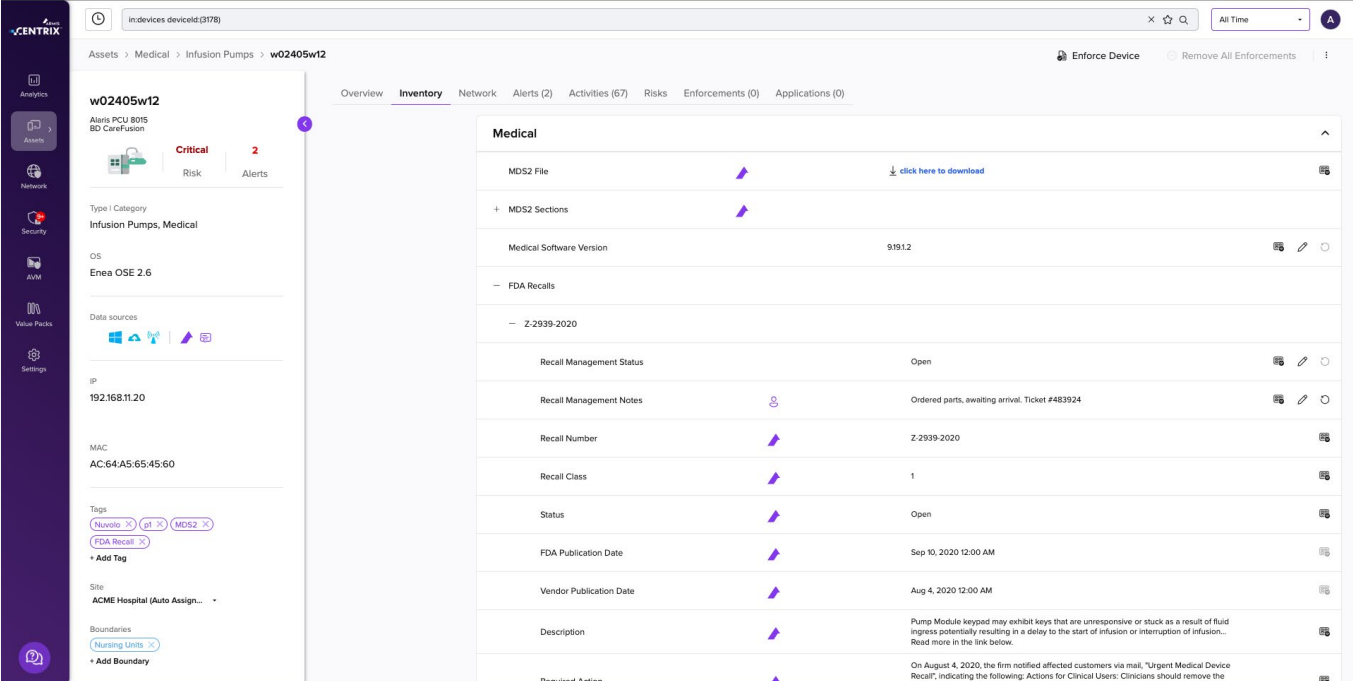
In addition to discovering and classifying a device, Armris calculates a risk score for every device based on factors like vulnerabilities, known attack patterns, and the behaviors associated with each device. Armris even factors in the clinical risk of devices, such as the presence or transmission of PHI, or the presence of active FDA recalls. This helps teams put risks into context, allowing them to quickly identify, assess, and remediate high-risk devices in a prioritized manner.

Like an agentless endpoint detection and response (EDR) tool for unmanaged and medical devices, Armris also continuously monitors the network for indicators of attack. When a device operates outside of its “known-good” profile, Armris issues an alert or triggers automated actions based on the policies you set for each device type. Everything from a misconfiguration to a policy violation to abnormal behavior, such as inappropriate connection requests or unusual software running on a device, can trigger an alert.



Armris Centrix™ discovers a wide range of hidden threats on your network.

- **Behavior** – Compares real-time device activity to established known-good baselines that are stored in the Asset Intelligence Engine. These are based on the historical behavior of the device, behavior of similar devices in your environment, and the behavior of similar devices in other environments.
- **Clinical risk** – Calculates clinical risk into the assessment of devices, facilitating enriched prioritization and strategic remediation.
- **Configuration** – Compares the configuration of each device to other devices within your environment, looking for anomalies.
- **Threat intelligence** – Utilizes premium threat intelligence and early warning alerts based on real world attack activity and patterns. Armris Centrix™ then uses exploit data insights to facilitate greater prioritization for vulnerabilities or risk factors that are most likely to be exploited within your environment, for proactive risk mitigation.



The screenshot displays the Armis Centrix web interface. On the left, a sidebar contains navigation icons for Analytics, Assets, Network, Security, AUM, Value Packs, and Settings. The main panel shows details for a device named 'w02405w12', identified as an 'Alaris PCU 8075 AD CardFusion'. It lists the OS as 'Enea OSE 2.6' and the IP as '192.168.11.20'. A 'Medical' section is expanded, showing a table of attributes:

Medical		
MDS2 File		click here to download
+ MDS2 Sections		
Medical Software Version		919.1.2
- FDA Recalls		
- Z-2939-2020		
Recall Management Status		Open
Recall Management Notes		Ordered parts, awaiting arrival. Ticket #483924
Recall Number		Z-2939-2020
Recall Class		1
Status		Open
FDA Publication Date		Sep 10, 2020 12:00 AM
Vendor Publication Date		Aug 4, 2020 12:00 AM
Description		Pump Module keypad may exhibit keys that are unresponsive or stuck as a result of fluid ingress potentially resulting in a delay to the start of infusion or interruption of infusion... Read more in the link below.
Remediated Action		On August 4, 2020, the firm notified affected customers via mail, "Urgent Medical Device Recall", indicating the following Actions for Clinical Users: Clinicians should remove the

Drill down capabilities enable you to quickly understand the specifics of device vulnerabilities.

Armis lets you see all the devices in your entire healthcare device ecosystem—medical, managed, unmanaged, and IoT. One solution for complete visibility and control.

Protect

Once Armis Centrix™ identifies malicious behavior on your network, or when a security policy violation is detected, you can respond manually or rely on automatic responses. For example, if a device transmits an abnormal amount of traffic or communicates with an unknown source, Armis Centrix™ can enforce an action to segment this device to prevent further risky behavior while maintaining essential functionality. Armis Centrix™ also monitors for and automatically takes preventive and corrective actions in response to zero-day vulnerabilities, threats, and exploits such as Log4j.

In addition to triggering protection policies based on malicious behaviors, Armis Centrix™ can also apply them proactively based on clinical risk or medical device properties. For example, proactively enforcing

medical device segmentation policies to reduce the attack surface from the onset of the device joining the network. Or, with respect to clinical activities, terminating external network communications when the platform detects unencrypted PHI destined for unsanctioned external networks.

In addition, your threat research team can also rely on Armis Centrix™ for help investigating the security incident. Armis continuously records information about the state and connections made by each device—managed or unmanaged—across your network so that when a security incident occurs, your security team can scroll back in time to see:

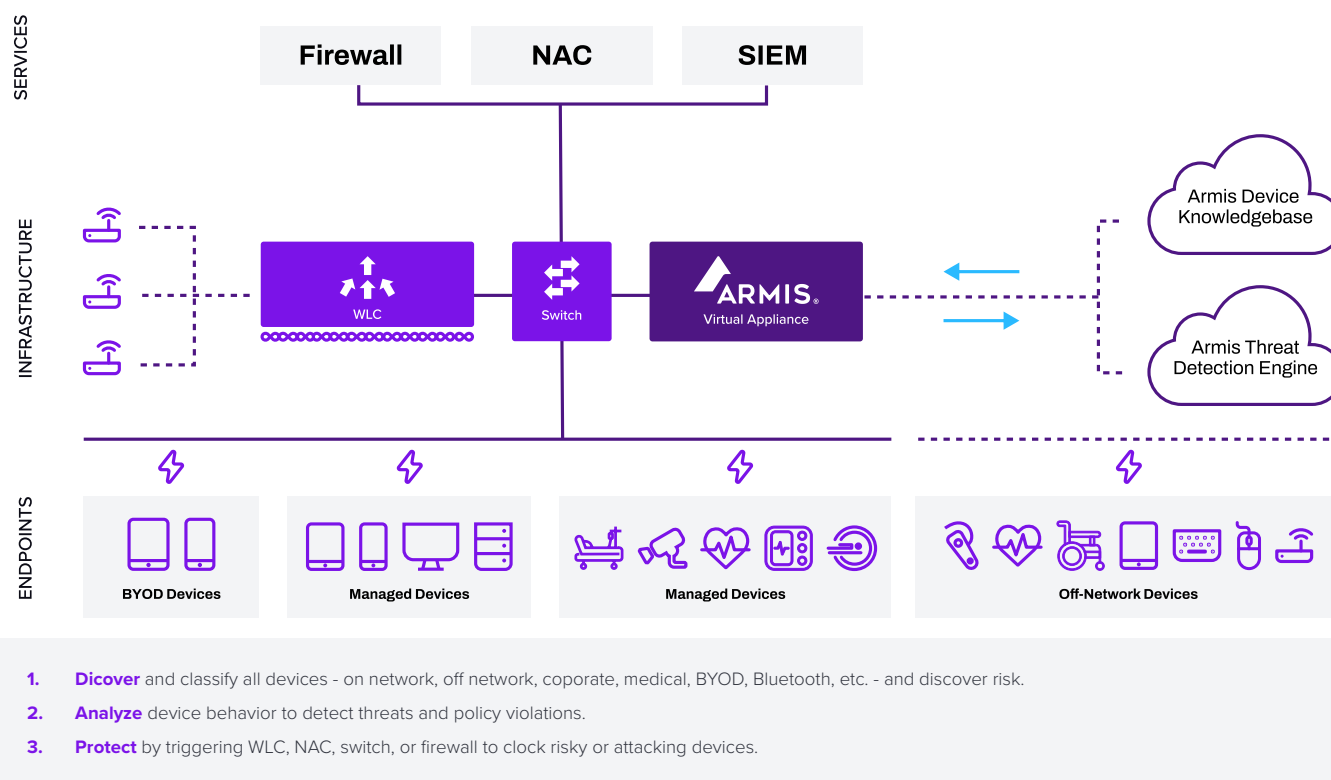
- The scope of the breach
- What communications occurred and over what protocols
- The amount of data transmitted
- Recent OS or application updates
- Abnormal traffic patterns
- Whether device locations have changed

For deeper analytics and threat mitigation, Armis Centrix™ integrates with existing security solutions to deliver a context-enriched, multi-data feed that provides a 360-degree view across an organization. These insights enable existing solutions to apply policies and mitigation against threats in a proactive manner to reduce the attack surface of the organization. The Armis platform integrates with a host of SIEMs, including Splunk.

Frictionless Installation

Most medical devices and other kinds of IoT devices can't support an agent. For this reason, Armis Centrix™ has been designed to deliver all its capabilities without the need for an agent. In addition, the platform does not increase latency or impact the production network in any way. In healthcare environments, this is critically important to the safe operation of biomedical devices.

Armis runs on your network as a virtual appliance, passively collecting information. It requires a simple user account on your existing wireless LAN controller. You also have the option to connect it to your wired network via a SPAN port and to your existing firewalls. Complete installation typically takes only minutes to a few hours, depending on the environment.



Armis Centrix™ is easy to install and integrate with your existing security and management tools.

Taking the next step

From the advent of the PC to the Internet to mobile devices to the cloud to IoT, history is a clear guide for the security risks associated with technological advances. And given that medical devices are built without security in mind, the risks to the patient journey are especially real and pressing. Today, hackers are targeting healthcare organizations more than any other industry.

New devices undeniably bring greater promise and innovation in the delivery of healthcare and the well-being of patients. But the security and clinical risks associated with managed and unmanaged devices are also all too real, so monitoring and securing them is key to ensuring the availability of patient care and minimizing risks to patient safety. The Armis platform underpins complete asset visibility and security. Your organization can install it within a couple of hours and have a complete inventory of devices in a week or less along with contextualized risk assessments of the entire environment and capabilities for automatically managing and reducing risk.

Without complete asset visibility there is no security, so now is the time to add IoT and medical device security to your comprehensive cyber security strategy.

Medical Devices That the Armis Platform Can Discover, Classify, and Display Within its Console

3M

AAEON Technology
 Abbott Diagnostics
 Abbott Optics
 Abbott Point of Care
 ACIST Medical Systems
 Acteon Group
 Advanced Sterilization Products
 Advanced Medical Information
 Advantage Pharmacy
 Aeroscout
 Alaris
 Alaris Medical Systems
 Alcon Laboratories
 Alpinion Medical Systems
 AmbiCom
 American Telecare
 Andon Health
 Applied Biosystems
 Applied Medical Technologies
 Arkray
 Avizia
 Axis Shield
 B Braun Melsungen
 Bang Olufsen Medcom
 Baxter Healthcare
 Beacon Medical
 Beckman Coulter
 Becton Dickinson
 BestCare Cloucal
 Bio logic Systems
 Bio Rad Lab

Biodevices

bioMerieux Italia
 Bionet
 BIOPAC Systems
 BioSoundLab
 Biospace
 Biotage
 BIOTRONIK
 BK Medical
 BL Healthcare
 BMT Medical Technology
 Boston Scientific
 BriteMED
 C8 MediSensors
 Calypso Medical
 Camtronics Medical Systems
 Canon
 CardioMEMS
 CardioNet
 Cardiopulmonary Corp
 CardioTek
 Care Everywhere
 CareCom
 CareFusion
 CareFusion Alaris Pump
 CarePredict
 Carestream Health
 CareTech
 CareView Communications
 Celectronic eHealth
 Centrak
 Cerner

CHG Hospital Beds

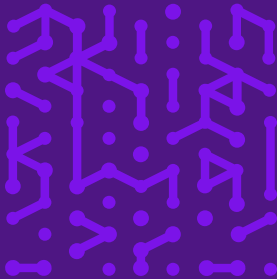
Cirtec Medical
 Cirtec Medical Systems
 CliniComp
 Cogent Healthcare Systems
 Colorado Med Tech
 Compex
 Compumedics
 Conmed Linvatec
 Convergent Bioscience
 Corometrics Medical Systems
 Criticare Systems
 Cutera
 Dainippon Pharma
 Danaher Motion Kollmorgen
 Datex Ohmeda
 DENTSPLY Gendex
 Diatek Patient Management
 Dictum Health
 Dixtal Biomedica
 Draeger
 Draeger Delta
 Draeger M300
 Dragerwerk
 Durr Dental
 Edwards Lifesciences
 Ellex Medical
 Essilor
 Fisher Paykel
 Fluke Biomedical
 Fresenius Medical Care

Fuji	Interacoustics	Mennen Medical
Fukuda Denshi	Intuitive Surgical	Micropoint Biotechnologies
Gambro Lundia	Invivo	Mindray
GE Healthcare	Ivoclar Vivadent	MIR
GE Medical	Ivy Biomedical	MOCACARE
GE Medical System	Johnson & Johnson Medical	Mortara Instrument
GEM Med	Jostra	NDS Surgical Imaging
Geringe	Karl Storz Imaging	Neural Image
Geringe IT Solutions	KaVo Dental	Nicolet Instruments
Geringe Sterilization	KeyMed	Nicolet Neuro
GN ReSound	Kodak Radiology	Nihon Kohden
Haag Streit	Kollmorgen	Nipro Diagnostics
Health Advice Monitors	Kollmorgen Corp	Nonin Medical
Health Hero	Kollmorgen Servotronic	Nova Biomedical
Health Life	Kontron Medical	Novo Nordisk
HealthStream	LABiTec	Olympus
Heart Force Medical	Laerdal Medical	Olympus Image Systems
HemoCue	Leica Biosystems	Omniceil
Heraeus Noblelight	Leica Microsystems	Omron Healthcare
Hitachi Aloka Medical	LI COR Biosciences	Onyx Healthcare
Hoana Medical	LifeSync	Optomedical Systems
Hologic, Inc.	LRE Medical	OrthoScan
Honeywell	Maquet	ORTHOsoft Zimmer CAS
Honeywell HomMed	Maquet Cardiopulmonary	Ortivus AB Medical
HORIBA Medical	Maquet CardioVascular	Oticon
Hospira	Maquet Critical Care	Pacific Biosciences
Huntleigh Healthcare	Maquet GmbH	PaloDEX
Imatron	Marconi Medical Systems	Palomar
Imricor Medical Systems	Masimo	Panasonic Healthcare
Indiana Life Sciences	Medav	Pharma Smart
InnerSpace	MedAvant Healthcare	Philips Analytical X Ray
Innomed Medical	Mediana	Philips Careservant
INSidE Technology	Medicis	Philips Healthcare PCCI
Integra Biosciences	Medicore	Philips Medical
Integra LifeSciences	Medison X Ray	Philips Oral Healthcare
Integrated Medical Systems	Medrad	Philips Patient Monitoring
Intel GE Care Innovations	Medtronic Diabetes	Philips Respirationics

Phonak Communications
Physio Control
Physiometrix
Planmeca Oy
Pointe Conception Medical
Power Medical Interventions
Progeny Midmark
Proteus Digital Health
Quantum Medical Imaging
Radiometer Medical
ResMed
Resurgent Health Medical
RF Surgical System
Robert Bosch
Roche Diagnostics
ScottCare
Secure Care
SenTec
Senticare
Shenzhen Lifesense Medical
Shimadzu
SHL Telemedicine
Siemens
Siemens Acuson Ultrasound
Siemens AG Healthcare Sector
Siemens Healthcare Diagnostics

Sigma
Sirona Dental Systems
Smiths Medical
SonoSite
Sonosite MicroMaxx Ultrasound
Soredex
Spacelabs Healthcare
Spectrum Medical Limited
Sphere Medical
St Jude Medical
Starkey Labs
Stratec Biomedical
Stryker
Sunol Molecular
Tecan Systems
Terumo
Thermo Fisher Scientific
Thoratec
Tiba Medical
Tokyo Boeki Medisys
Toyo Medic
tPlus Medical
Trendsetter Medical
Tunstall Healthcare
Valtronic
Varian Medical Systems

Versamed
Verto Medical
VIASYS Healthcare
Vigil Health Solutions
VitalCare
Vocera
Welch Allyn
West Com Nurse Call
Widex
Zimmer Elektromedizin
Zoe Medical
ZOLL Lifecor



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